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EduOS Cameroon — Investment-Grade Document Pack

This folder upgrades the EduOS Cameroon vision suite into materials that can survive financier appraisal and vendor procurement. Where the vision volumes describe *what and why*, these documents commit to *how much, how measured, and how built* — with numbers, sources, and testable requirements.

The pack

Doc	Title	What it answers
01	Baseline Data Annex	The verified evidence base: 7.3M learners, 22,000+ primary schools, 12:1→1.5:1 textbook ratio history, 26% rural electrification, budgets — every figure sourced (World Bank API, MINEDUB/MINESEC, DataReportal), gaps flagged for appraisal
02	Costed Budget & 5-Year TCO	11.62 bn FCFA (US\$20.2M) phased programme, bottom-up unit costs, 1.22 bn FCFA/yr steady-state recurrent (= 0.13% of the two ministries' budgets), financing strategy
03	Economic & Financial Analysis	CBA over 10 years: central-case NPV +9.5 bn FCFA, BCR 1.65, EIRR ~22%, payback Y6; NPV-positive if just 2.1% of the textbook flow is recovered
04	FRS — National Textbook Registry	The buildable module spec: NTID/NCID identifier scheme, normative data model, state machines, 40+ numbered requirements with acceptance criteria, API contracts, offline sync rules, NFRs, traceability matrix
05	M&E Framework & Results Matrix	Theory of change, indicators with baselines/targets by year, independent verification design, phase-gate disbursement linkage
06	Risk Register & Mitigation Plan	Scored register; programme-critical risks (dual-ministry governance, connectivity reality, post-donor financing, vendor lock-in) each with budgeted mitigations
07	FRS — National School Registry	The foundational registry: NSID scheme, duplicate detection with measured precision/recall, enrolment validation workflow, carte scolaire import pipeline, GPS verification campaign
08	FRS — Warehouse & Distribution (NWIDMS)	The custody engine: double-entry stock ledger, chain-of-custody shipment state machine, automatic discrepancy cases (zero silent variance), redistribution engine, emergency mode
09	Tech Stack & Architecture Decisions	ADR-format decisions: modular monolith with DDD bounded contexts, PostgreSQL/Laravel/Flutter open-source stack, API-first with APIs as national digital public infrastructure, tiered rugged-device strategy — plus an honest open-gaps register
10	Pre-Build Readiness Checklist	The gate to development: blocking vs parallel items with owners — governance decree, financing, procurement route, FRS sign-off,

Doc	Title	What it answers
		Phase-0 engineering (sync engine, offline auth, OpenAPI), data/legal, pilot baselines
11	Master Implementation Plan	The single picture: what is being built (one-page system view + the five guarantees), the two-clock (T0/school-year) timeline to T0+60m, four-track Phase I, the dependency spine including devices and print seasons, gate-failure protocol, and what “done” means
12	E&S Safeguards, Inclusion & Conflict Sensitivity	ESF-mapped safeguards: the e-waste/battery reverse-logistics plan, gender & disability inclusion requirements (INC-01..04), do-no-harm conflict analysis for NW/SW/Far North, stakeholder engagement and grievance mechanism
13	Procurement Plan (skeleton)	Eleven procurement packages with methods, values, and MIP-keyed timing — including the class-spec device tender with wave options and the e-waste treatment contract
14	Market & Engagement Strategy	The go-to-market: the ERSP-closing window (Dec 2026), TTL/GPE engagement sequence with deadlines, Digital Public Good registration play, the verified six-country replication market, and Opesware’s open-source business model

Cross-references are live: the Budget funds the Risk mitigations, the M&E gates the Budget’s disbursement phases, the Economic Analysis consumes the Baseline Annex, and the FRS trio (04, 07, 08 — the full Phase-I module scope) defines what the M&E measures.

Data-gap status — all four closed (BDA §6)

- ~~Secondary school count~~ — **4,131 establishments** (MINESEC Annuaire 2017/18, parsed from source); refresh to 2024/25 tables at appraisal.
- ~~Textbook unit costs~~ — **US\$6.25 → US\$2.90 (≈1,750 FCFA)** verified; 11M+ books distributed under PAREC.
- ~~PETS-III leakage~~ — **~30% value leakage measured** on school supply packages + 3–6 month delays; now the anchor evidence of the economic case (break-even needs recovery of 1/14th of the measured loss rate).
- ~~Carte scolaire dataset~~ — verified that **no open machine-readable school-GPS dataset exists** for Cameroon; the NSR seeding strategy (FRS-NSR §9) is designed around that reality, and the registry itself becomes the country’s first authoritative open school dataset.

Roadmap to development

1. **Now → appraisal:** ministry review of the three FRS documents; inter-ministerial governance decree drafted (Risk R1 precondition); refresh BDA with 2024/25 annuaire tables and ERSP ISR spend figures.
2. **Before build (procurement stage):** produce OpenAPI 3.1 files from FRS §7 of each module (deliverables D-NTR-API, D-NSR-API, D-NWD-API); assemble the bidding pack (FRS trio + acceptance criteria + NFRs).
3. **Then development:** Phase I build against FRS acceptance criteria, 500-school pilot, baselines frozen (M&E §3.2), Phase II gate.